Sequences — problem 9.58

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The following problems treat the number a_n of ways to tile a $k \times n$ rectangle by various smaller tiles. A solution is here a recurrence for a_n.

Let a_n be the number of ways to tile a $2 \times n$ rectangle by 2×1 dominoes. (a) Find a_n. (b) Find the number of symmetric and distinct tilings.